

Setup and Deployment Guide

Install, configure, validate, and publish DevPulse Version 1 safely.

Release	Version 1
Platform	.NET 9 Blazor Server
Audience	Developers, operators, reviewers, and project maintainers
Status	Presentation and deployment package

Observe. Test. Inspect. Plan. Focus.

CREDITS

Project creators

Add the names of the DevPulse student team and project advisor before sharing or presenting this document.

Role	Name
Student 1	_____
Student 2	_____
Student 3	_____
Advisor	_____

Where to edit the names

Open page 2 in Microsoft Word. Click the blank line in the Name column, delete the underscores, and type the correct name. Repeat this for all three students and the advisor.

Quick start

A clean local launch needs only the .NET 9 SDK. Spotify, browser audits, alerts, and private-network checks are optional integrations.

Five-minute local launch

1. Open a terminal in the repository root.
2. Restore and build the application.
3. Run the HTTP launch profile.
4. Open `http://127.0.0.1:5214` in a browser.
5. Confirm that `/healthz` returns a healthy result.

```
dotnet restore
dotnet build
dotnet run --launch-profile http
```

What works without credentials

Overview, network calculators, local inspectors, Focus Lounge wellness tools, and the configured diagnostics workspace can run without Spotify credentials. Production diagnostics are disabled unless the owner explicitly enables them.

Recommended first demonstration

- Open Overview and explain that metrics describe the DevPulse host.
- Calculate `192.168.10.25/24` in Network Calculator.
- Run a Website Audit against a site you own.
- Finish in Focus Lounge and optionally connect Spotify.

Requirements and repository

Install the components that match the capabilities you plan to demonstrate.

Component	Required	Purpose
.NET 9 SDK	Yes	Build and run the Blazor Server application
Visual Studio 2022 17.12+	Optional	IDE support for .NET 9
Git	Recommended	Source control and deployment integration
Node.js plus Chromium	Optional locally	Browser audit and Lighthouse-style checks
Docker	Optional	Container validation and hosted deployment
Spotify Premium	Optional	Spotify Connect controls and browser playback

Important project files

File or folder	Purpose
MiniProject_Everything_1.csproj	.NET 9 target and package references
Program.cs	Services, authentication, routes, security middleware
appsettings.json	Safe defaults and non-secret configuration
Components/	Razor pages, layout, and interface components
Service/	Diagnostics, Spotify, networking, audit, and storage logic
scripts/	Browser audit and verification automation
Dockerfile	Production image on port 10000

Do not change the target framework

DevPulse Version 1 targets net9.0. Install .NET 9 and keep the project on .NET 9 unless a planned migration is tested separately.

Configuration and secret handling

Keep operational settings readable, but keep every password, token, client secret, and private endpoint outside Git.

Configuration precedence

1. appsettings.json provides safe committed defaults.
2. appsettings.Development.json can provide non-secret development overrides.
3. .NET User Secrets provide local credentials.
4. Environment variables or a mounted secret file provide hosted credentials.

Local secrets

```
dotnet user-secrets set "Spotify:ClientId" "YOUR_CLIENT_ID"  
dotnet user-secrets set "Spotify:ClientSecret" "YOUR_CLIENT_SECRET"  
dotnet user-secrets set "Admin:Password" "A-unique-password-of-at-least-12-characters"
```

Hosted environment variables

```
Spotify__ClientId=YOUR_CLIENT_ID  
Spotify__ClientSecret=YOUR_CLIENT_SECRET  
Admin__Password=YOUR_LONG_UNIQUE_PASSWORD  
Diagnostics__Enabled=false
```

Leak prevention

Never place real credentials in appsettings.json, source code, screenshots, test files, Git history, or presentation material. Rotate a credential immediately if it has ever been committed.

Persistent cryptographic data

- Mount a persistent data directory at /app/data in production.
- Keep data-protection keys stable across container replacement.
- Spotify token sessions are encrypted on the server; the browser receives only an opaque session identifier.

Spotify integration

Spotify is optional. The rest of DevPulse remains usable when the integration is not configured.

Spotify developer dashboard

1. Create or open your application in the Spotify Developer Dashboard.
2. Add your account and any testers allowed by the application mode.
3. Register the exact redirect URI. Spotify requires an exact character-for-character match.
4. Store the client ID and client secret using User Secrets or hosted secret variables.
5. Restart DevPulse, open Focus Lounge, and connect Spotify.

Environment	Redirect URI
Local HTTP profile	http://127.0.0.1:5214/signin-spotify
Render production	https://devpulse-project-sng7.onrender.com/signin-spotify
Custom domain	https://YOUR-DOMAIN/signin-spotify

Redirect mismatch

Update both the Spotify dashboard and the deployed application URL. Do not add or remove a trailing slash, change localhost to 127.0.0.1, or switch HTTP and HTTPS unless the registered value changes too.

Playback behavior

- Spotify Connect follows the active phone, desktop, speaker, or browser device.
- Selecting Continue on this device transfers playback while preserving play or pause state.
- Browser audio requires Premium and a user gesture to enable browser playback.
- Progress reconciles with Spotify about every four seconds because Spotify provides no playback webhook.

Diagnostics, allowlists, and private targets

Network tools run from the DevPulse host, not from the visitor's browser. Owner-controlled allowlists define what the host may contact.

Enable diagnostics intentionally

```
Diagnostics__Enabled=true
Diagnostics__FolderRoot=D:\approved\logs
```

Setting	Example	Effect
Operations__ApprovedHosts__0	127.0.0.1	Allows ping, DNS, TLS, and TCP checks for one host
Operations__ApprovedPorts__0	443	Adds an allowed TCP port
Operations__ApprovedUrls__0	https://api.github.com	Adds a URL suggestion or exact trusted exception
Operations__ApprovedLogFiles__0	/var/log/app.log	Allows the log tailer to read one file
Operations__Databases__0__Kind	TCP or Redis	Selects TCP accept or unauthenticated Redis PING

Private addresses

Local, private, link-local, and reserved addresses are blocked for arbitrary input. Add an exact owner-approved host when you intentionally test your own LAN or local service. A Render container still cannot reach a private computer behind your router.

Bounded execution

- Load tests are capped at 25 requests and five concurrent workers by default.
- Redirects are disabled for diagnostic HTTP clients.
- Response previews, request bodies, timeouts, scan ports, and crawl resources have explicit limits.

Website, API, DNS, and alert setup

These modules are useful for demonstrating DevPulse against systems you own without changing source code.

Website Audit

```
WebsiteAudit__DefaultUrl=https://example.com/  
WebsiteAudit__MonitoredUrls__0=https://example.com/  
WebsiteAudit__BrowserEnabled=true
```

- Use a public site you own or are authorized to test.
- The server performs crawl, header, JSON, accessibility, visual, and optional browser checks.
- Scheduled uptime checks run at the configured interval and create incidents when state changes.

API Collection Runner

- GET and HEAD may target validated public endpoints.
- POST, PUT, PATCH, and DELETE require the exact endpoint in Operations:ApprovedUrls.
- Authorization values and custom headers remain temporary and are not written to disk.

DNS and Email Inspector

- Inspect A, AAAA, MX, SPF, DMARC, DKIM, and CAA records for public domains.
- Use selectors from your mail provider when testing DKIM.
- Results are diagnostic evidence, not a guarantee of message delivery.

Alerts

```
Alerts__MemoryMb=1024  
Alerts__DiskUsedPercent=90  
Alerts__RequestLatencyMs=2000  
Alerts__Smtplib__Password=STORE_AS_A_SECRET
```

Local validation

Run the same checks before presenting or deploying. A successful page load alone is not enough release evidence.

Build and format

```
dotnet restore
dotnet build -c Release
dotnet format --verify-no-changes
```

Regression and smoke checks

```
dotnet run --project tests/DevPulse.Tests/DevPulse.Tests.csproj
powershell -ExecutionPolicy Bypass -File scripts/http-smoke.ps1
```

Version 1 verification baseline

The final Version 1 validation recorded 79 of 79 automated checks passing, a Release build with zero warnings and zero errors, a successful formatting check, and a successful published HTTP smoke suite.

Manual presentation checks

- Open every navigation item at desktop width.
- Run one approved network operation and one website audit.
- Upload only non-sensitive sample files to the inspectors.
- Connect Spotify, browse created and saved collections, and confirm the active device behavior.
- Confirm administrator login and logout without enabling process termination.

Docker and Render deployment

The repository Dockerfile publishes a .NET 9 application, installs the browser-audit runtime, and runs as the non-root app user.

Container contract

Item	Value
Internal HTTP port	10000
Health endpoint	/healthz
Persistent data path	/app/data
Runtime user	app (non-root)
Browser runtime	Node.js 24 plus Chromium

Render setup

1. Create a Web Service from the repository and choose Docker as the runtime.
2. Use port 10000 and configure the health check path as /healthz.
3. Add Spotify and administrator credentials as secret environment variables.
4. Mount persistent storage at /app/data if Spotify sessions and telemetry must survive replacement.
5. Register the final HTTPS Spotify callback and configure trusted proxy addresses or networks.
6. Deploy one instance unless shared session and telemetry storage are introduced.

Reverse proxy safety

Trust only the proxy ranges supplied by the hosting platform. Do not configure forwarded headers to trust every network, because OAuth depends on the external HTTPS scheme being trustworthy.

Production security checklist

DevPulse contains diagnostics that can reveal host and network information. A public demo needs stricter access controls than a local classroom demonstration.

Control	Required action
Diagnostic exposure	Keep disabled or place behind authenticated access
Administrator access	Use a unique password of at least 12 characters
Process termination	Keep disabled unless explicitly reviewed
Private target checks	Allowlist only systems you own or are permitted to test
Secrets	Use environment variables or a mounted secret file
Data volume	Persist and protect encryption keys and application state
Public reputation	Resolve Search Console and Safe Browsing warnings before promotion
API screenshots	Never capture tokens, keys, or authorization headers

Current public launch checkpoint

Before presenting the public Render URL broadly, complete the Google Search Console or Safe Browsing review and place diagnostic routes behind authenticated access or a private gateway.

Safe default

```
Diagnostics__Enabled=false  
Admin__AllowProcessTermination=false
```

Troubleshooting

Start with the symptom, confirm which machine is running the check, and then review the relevant configuration boundary.

Symptom	Likely cause	Resolution
redirect_uri mismatch	Registered callback differs from the request	Make the Spotify dashboard URI exactly match scheme, host, port, path, and slash
Diagnostics are disabled	Production default is off	Enable Diagnostics only for an intentionally protected deployment
Storage shows Render disks	Checks run on the server host	Use a local deployment or future remote agent for workstation storage
Private IP is blocked	SSRF protections reject private input	Add an exact approved host and run DevPulse where that network is reachable
No Spotify sound	Playback remains on another device or browser audio is not activated	Select an active device or explicitly enable browser playback with Premium
Created playlist is missing	OAuth scopes are stale or the item is not owned/collaborative	Disconnect, reconnect, and verify playlist ownership and scopes
Chrome warns about the site	Safe Browsing reputation or diagnostic content	Stop broad sharing, review Search Console, secure routes, and request review
Browser audit unavailable	Node or Chromium path is missing	Install the optional runtime or use the Docker image

Release and handoff checklist

Use this page before a presentation, deployment, or handoff to another maintainer.

- The repository builds with the .NET 9 SDK.
- No real credential is present in committed files or screenshots.
- The public hostname and Spotify callback match exactly.
- The /healthz endpoint is healthy after deployment.
- Persistent storage is mounted when encrypted sessions or telemetry must survive restart.
- Diagnostics are disabled or protected by authenticated access.
- Only authorized hosts, ports, URLs, log files, and database endpoints are allowlisted.
- The Release build, format check, regression executable, and HTTP smoke suite pass.
- A manual walkthrough has been completed on the device used for the presentation.
- Google Safe Browsing or Search Console status is clear before public promotion.

Useful endpoints

Path	Purpose
/	Overview
/healthz	Platform health check
/focus-lounge	Focus tools and Spotify
/website-audit	Website assurance
/api-runner	Bounded API collections
/admin	Restricted administration

Version 1 is ready to present

The application and release package are technically validated. The remaining public-launch work is operational: protect diagnostics and complete the browser reputation review.